

Week 6 Update — COVID-19 Global Data Analysis

Shrey Jain • Interactive Plotly charts for case & death trends • Prepared for Neelam Gupta

Weekly update

This week I built four interactive Plotly charts for case and death trends, each exported as a standalone HTML file that opens in any browser and reproducible from the notebook. The images below are static previews — the live versions support hover, zoom, legend toggles, dropdowns, a range slider, and animation.

Built:

- **Global cases vs deaths** — dual-axis 7-day-smoothed trends with a range slider and 3m / 6m / 1y / All quick-range buttons.
- **Country explorer** — a metric dropdown (new cases, new deaths, cumulative cases, cumulative deaths, all per million), a linear/log toggle, legend isolation, and a range slider.
- **Continent small multiples** — faceted new-cases-per-million panels with independent axes.
- **Animated world map** — a monthly choropleth of new cases per million with a play button and time slider.

Chart previews

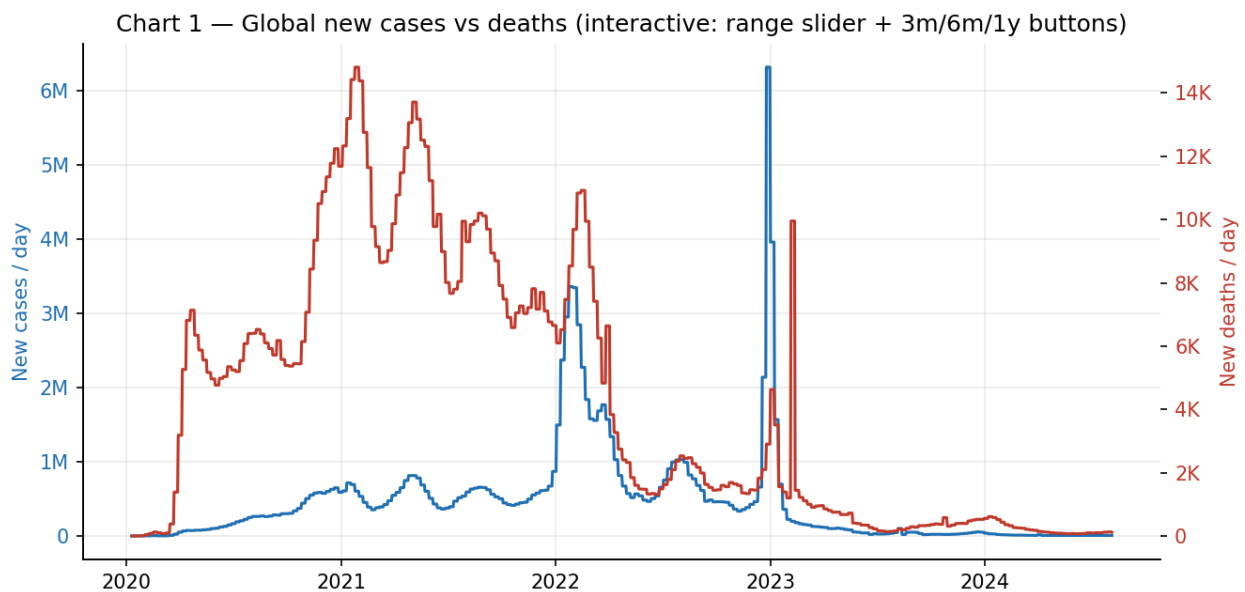


Chart 1 preview — global new cases vs deaths. Live version adds a range slider and quick-range buttons.

Chart 2 — Country explorer (interactive: metric dropdown, log toggle, legend, slider)

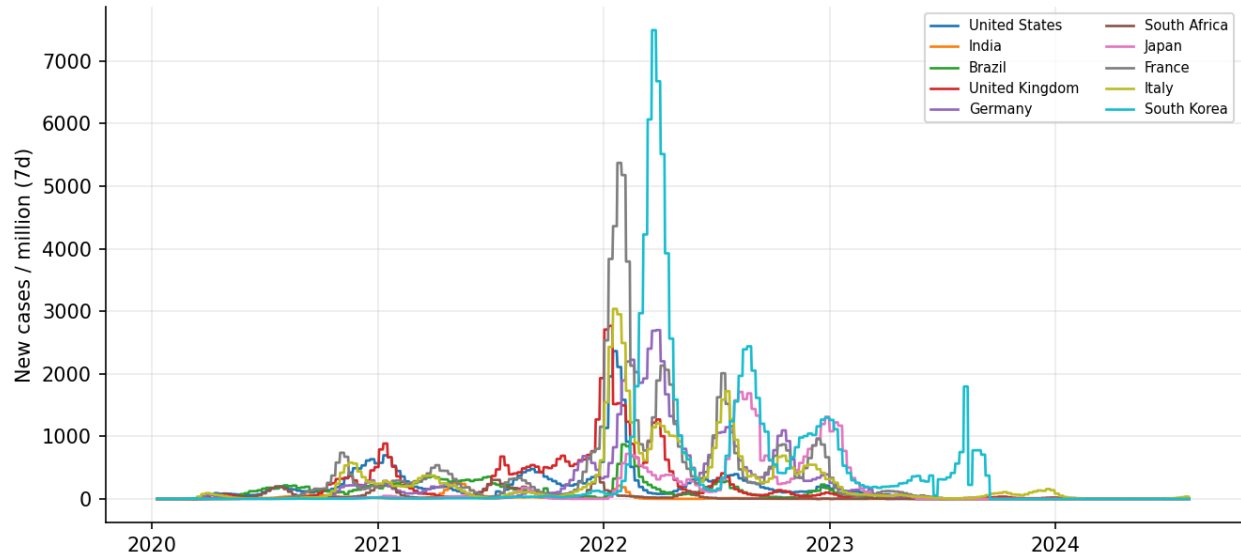


Chart 2 preview — country explorer (default metric shown). Live version adds a metric dropdown, log/linear toggle, and legend isolation.

Chart 3 — New cases/M by continent, faceted small multiples (interactive hover)

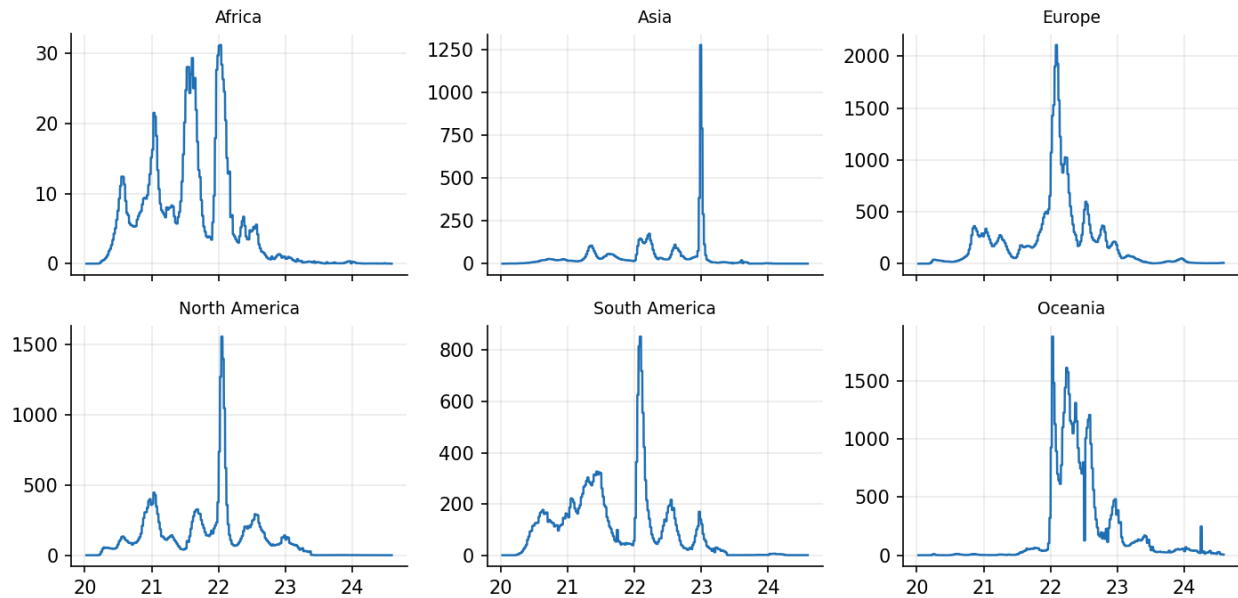
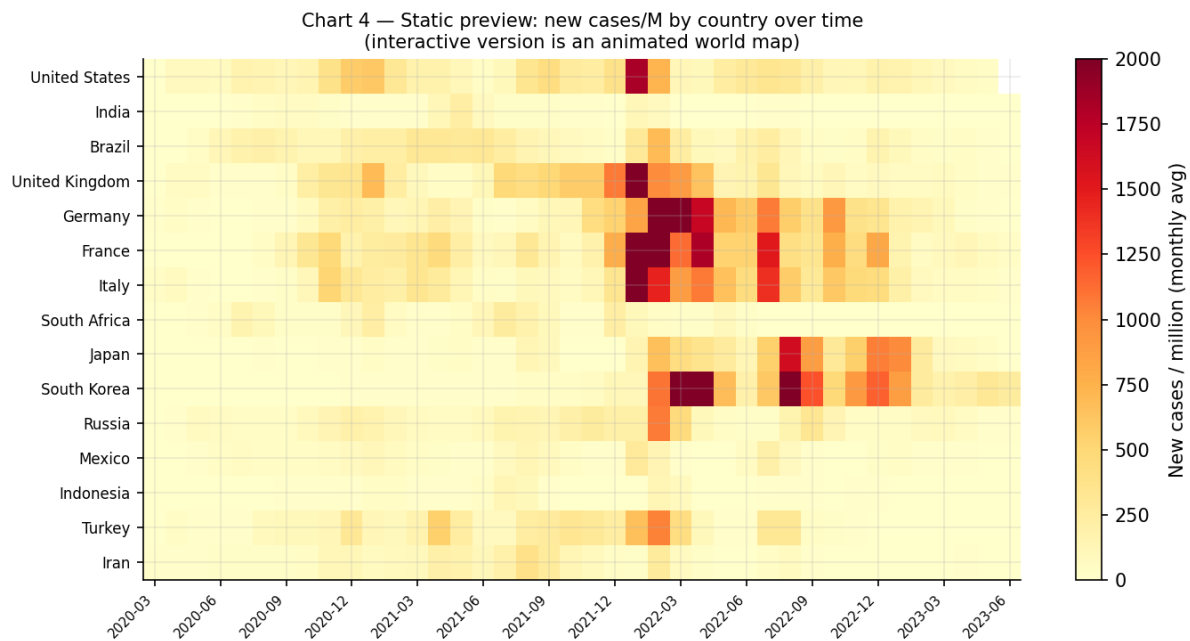


Chart 3 preview — continent small multiples with independent y-axes.



Work artifacts & data sources

Repository	github.com/shreyjain99/covid19-global-analysis
Week 6 folder	github.com/shreyjain99/covid19-global-analysis/tree/main/week%206
Interactive charts	charts/chart1–chart4 (.html) — open in any browser
Week 6 notebook	week6_interactive_charts.ipynb (renders all four inline)
Data source	Our World in Data COVID-19 dataset (github.com/owid/covid-19-data)

Reproducibility (how to re-run)

- `pip install -r requirements.txt`.
- Open `week6_interactive_charts.ipynb` and Run All — it downloads the dataset, renders all four charts inline, and writes the HTML copies into `charts/`.
- Or: `python scripts/build_charts.py` to regenerate the HTML charts, and `python scripts/build_week6_report.py` to rebuild this PDF.
- To view a chart, open any file in `charts/` in a web browser.

Risks / blockers & support needed

- **No blockers to delivery.** The week's objective is met and reproducible.
- **Format note:** the charts are interactive HTML, so this PDF shows static previews; the live interactivity is in the .html files and the notebook.

- **Data note:** the animated map caps new cases/M at 2,500 so later spikes don't wash out the colour scale, and trends reflect each country's reporting cadence.
- **Support:** confirm the preferred delivery format (standalone HTML files vs an embedded dashboard) if these are to be shared more widely.